SPHOORTHY ENGINEERING COLLEGE | *(UGC Autonomous — NAAC Grade A)*

Python for AI/ML: NumPy, Pandas, Matplotlib & Scientific Computing

Department: CSE AIML	**Course / Subject:** Python for Data Science (AI401PC)
Semester: Semester 4	**Regulation:** Autonomous (SR23 / SR24)

Course Overview & Key Focus: Vectorized computations with NumPy, Data wrangling using Pandas DataFrames, Data visualization with Seaborn/Matplotlib, and Scipy optimization tools.

Unit 1: Python Advanced Foundations

List comprehensions, Lambda functions, Generators, Decorators, Object-Oriented Python classes, and Exception handling paradigms.

Unit 2: NumPy for Numerical Arrays

N-dimensional array indexing, Slicing, Broadcasting rules, Linear algebra matrix decomposition (SVD, Eigenvalues), and Vectorized operations.

Unit 3: Pandas Data Manipulation

Series and DataFrames, Missing value imputation, GroupBy aggregations, Merging/Joining datasets, and Time-series data resampling.

Unit 4: Statistical Data Visualization

Matplotlib subplots, Seaborn heatmaps, Histograms, Box plots for outlier detection, Pairplots, and Plotly interactive charts.

Unit 5: Data Preprocessing for AI

Feature scaling (StandardScaler, MinMaxScaler), One-Hot Encoding, Label Encoding, Train-Test split stratification, and Dimensionality Reduction (PCA).